UF AI DAYS · OCT. 27, 2026

AI Research, Teaching, and Learning Day

Restoring the Milky Way

Scaling AI-assisted light pollution planning across North Central Florida



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Research Software Engineer III, University of Florida College of Education

Photo: Milky Way and urban skyglow over Paynes Prairie, 2022. Jay Rosen

About me: from light art to light pollution

- **Light art, 15 years**: light, mirrors, and increasingly technological media
- **Before UF**: founder and CEO, Space Age Mining Corp. (HPC)
- **At UF**: Research Software Engineer III, College of Education
- **Astrophotography since 2020**: photos on state park websites
- **NASA Space Apps**: 2× Kennedy Space Center winner, global nominee, judge
- **Outside UF**: Board of Directors, Alachua Astronomy Club
- Civic service on **City of Gainesville** and **Alachua County** boards



Full-spectrum self-portrait at the Chiefland Astronomy Village, April 2024. Photo: Jay Rosen

THE PROBLEM

Skyglow travels: one city's lights wash out a whole region



Paynes Prairie · Bortle 4, Gainesville dome. Gainesville's skyglow and Comet NEOWISE from Paynes Prairie, July 2020. Photo: Jay Rosen



Stephen C. Foster State Park, GA · Bortle 2. International Dark Sky Park, October 2020. Photo: Jay Rosen

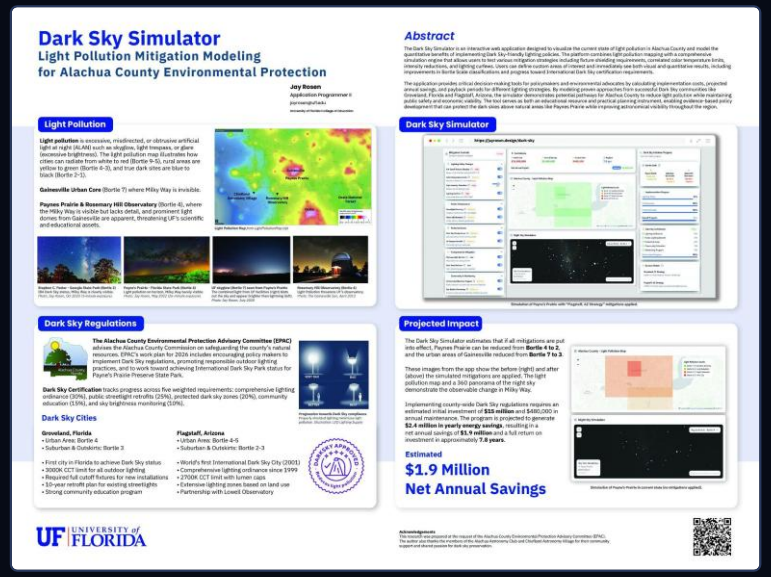
- Light scattered by air and haze brightens the sky **tens of km** away
- Harms wildlife, sleep and health, and wastes energy

- Threatens UF's dark sites: **Rosemary Hill Observatory** and **Chiefland Astronomy Village**
- **Preventable**: shield, aim, warmer color, less brightness, dim late at night

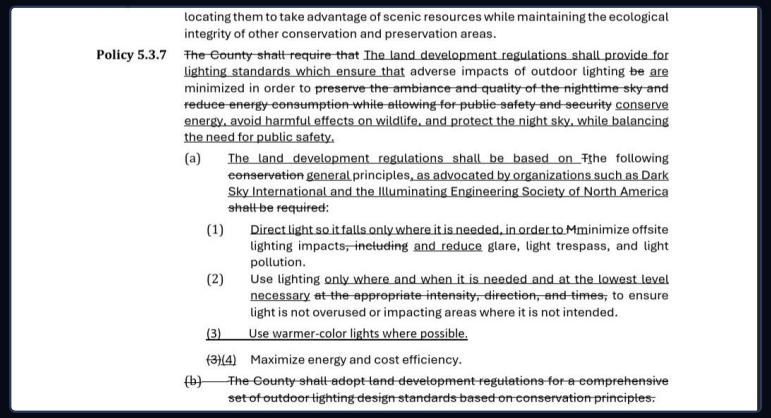
BACKSTORY

From astrophotography to county policy

- **2020–25** Photographing the night sky under Gainesville's light dome
- **Sep 2025** Public comment at Alachua County's advisory committee (EPAC)
- **Oct 2025** v1 simulator, built in a month: mitigations' economic and night-sky impact; debuted at UF Astraeus Space Research Day
- **Jan 2026** EPAC votes unanimously to work toward dark-sky policy
- **Feb–May 2026** Policy 5.3.7 amendment heard and sent to the state
- **Sep 22, 2026** County Commission **adopts** it, unanimously
- **2026** Expanding into **Levy County**, home to Rosemary Hill and the Chiefland Astronomy Village
- **Next** UF Astronomy (Dr. Elizabeth Lada) asks where a future observatory should go



v1 poster, UF Astraeus Space Research Day, Oct 2025



Policy 5.3.7 The County shall require that The land development regulations shall provide for lighting standards which ensure that adverse impacts of outdoor lighting ~~be~~ are minimized in order to ~~preserve the ambiance and quality of the nighttime sky and reduce energy consumption while allowing for public safety and security~~ conserve energy, avoid harmful effects on wildlife, and protect the night sky, while balancing the need for public safety.

(a) The land development regulations shall be based on ~~T~~the following ~~conservation~~ general principles, as advocated by organizations such as Dark Sky International and the Illuminating Engineering Society of North America ~~shall be required~~:

(1) Direct light so it falls only where it is needed, in order to ~~M~~minimize offsite lighting impacts~~, including~~ and reduce glare, light trespass, and light pollution.

(2) Use lighting only where and when it is needed and at the lowest level necessary ~~at the appropriate intensity, direction, and times,~~ to ensure light is not overused or impacting areas where it is not intended.

(3) Use warmer-color lights where possible.

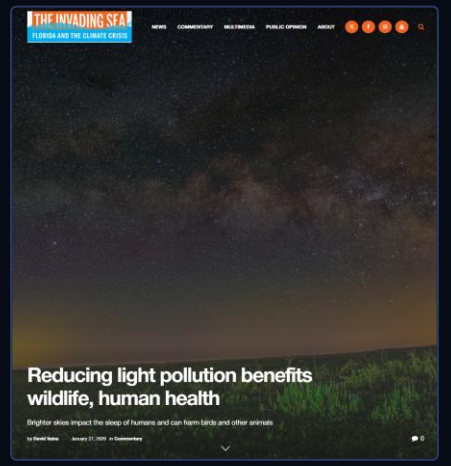
~~(3)~~(4) Maximize energy and cost efficiency.

~~(b) The County shall adopt land development regulations for a comprehensive set of outdoor lighting design standards based on conservation principles.~~

Policy 5.3.7 language, Feb 2026



UF College of the Arts, Mar 2026

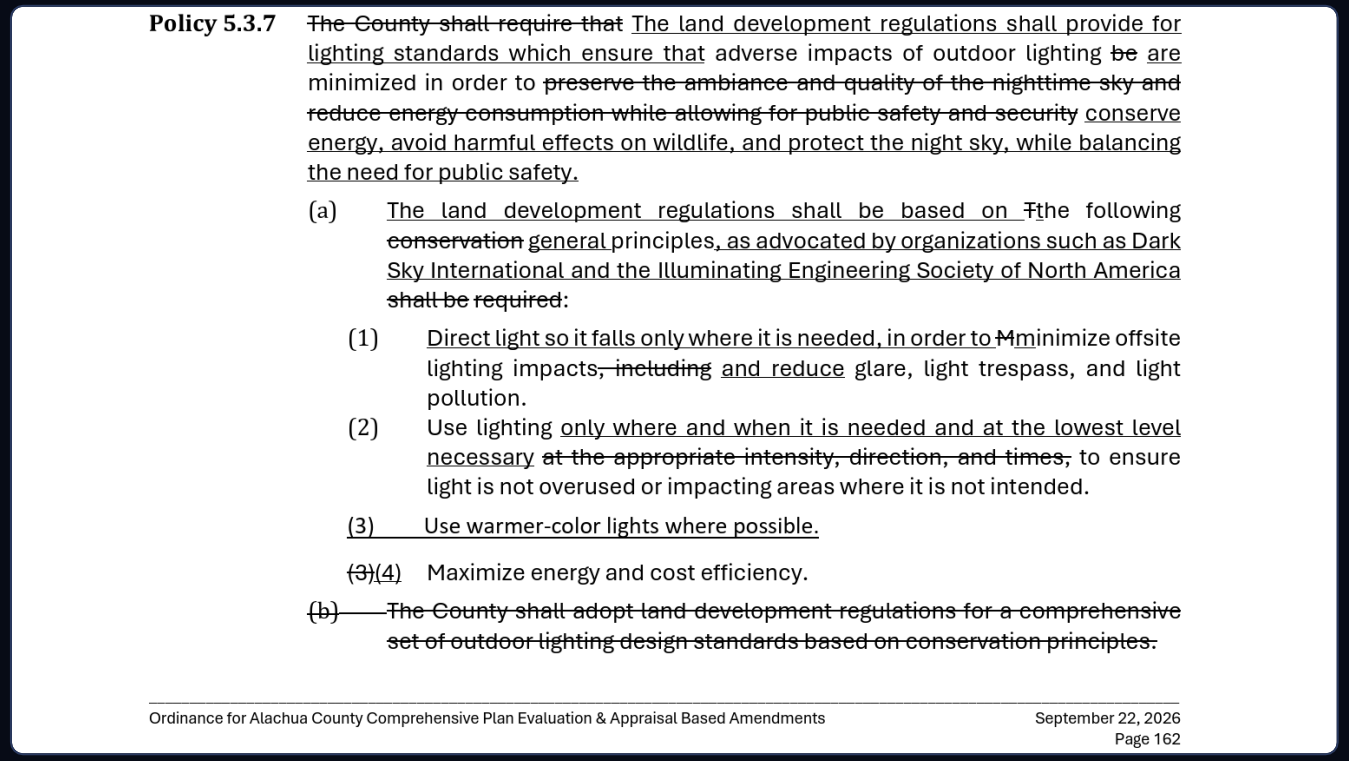


The Invading Sea, Jan 2026

BACKSTORY

September 2026: dark-sky principles adopted

- Alachua County's Comprehensive Plan update, **Policy 5.3.7** (Conservation and Open Space Element)
- Lighting standards must **conserve energy, avoid harm to wildlife and protect the night sky**, balanced with public safety
- Based on principles from **Dark Sky International** and the **Illuminating Engineering Society**
- Light only where and when needed, at the lowest level; **warmer colors**; energy and cost efficiency
- Each one is a control in the simulator: shielding, dimming, the 3000 K cap, costs
- Next: the land development code; in effect once the state finds it in compliance



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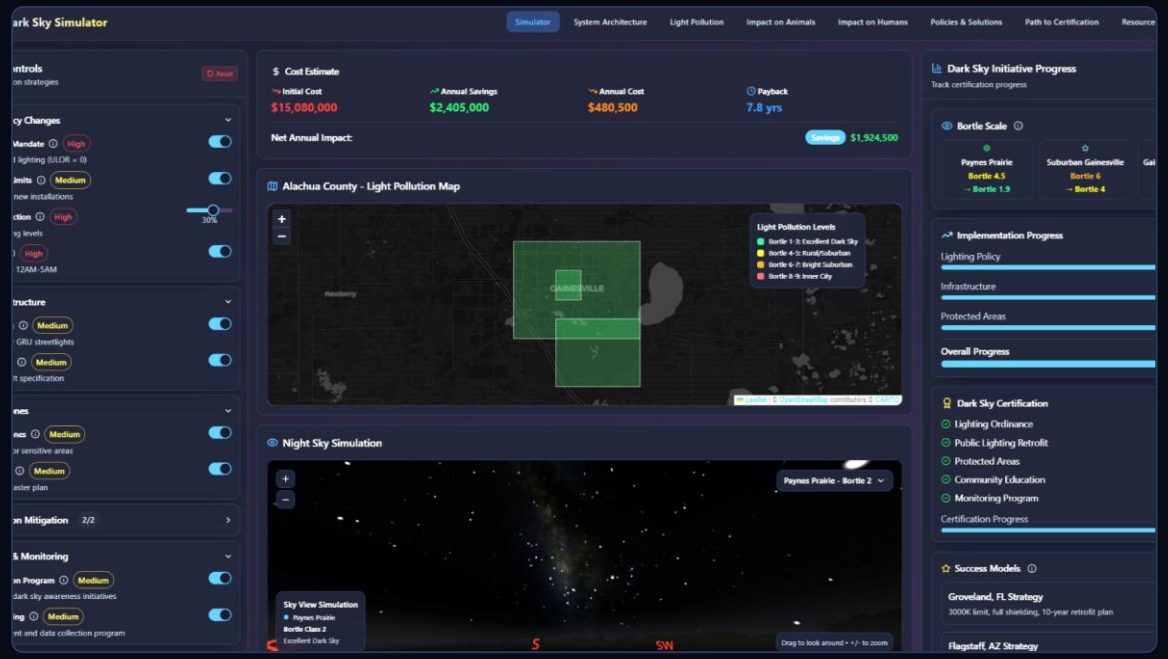
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Ordinance for Alachua County Comprehensive Plan Evaluation & Appraisal Based Amendments — September 22, 2026 Page 162

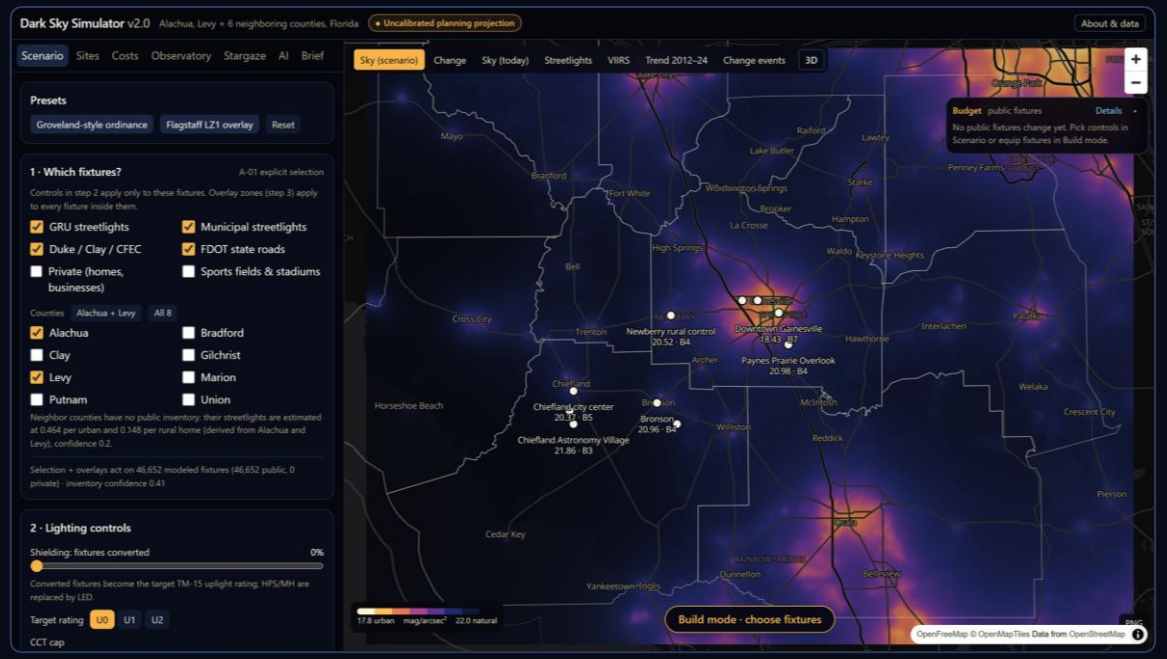
Policy 5.3.7 as adopted: underlined text added, struck text removed. Alachua County ordinance amending the Comprehensive Plan (Evaluation & Appraisal amendments), September 2026, p. 162.

WHY A VERSION 2

v1 told a story. v2 has to answer “How do you know?”



v1 · 2025



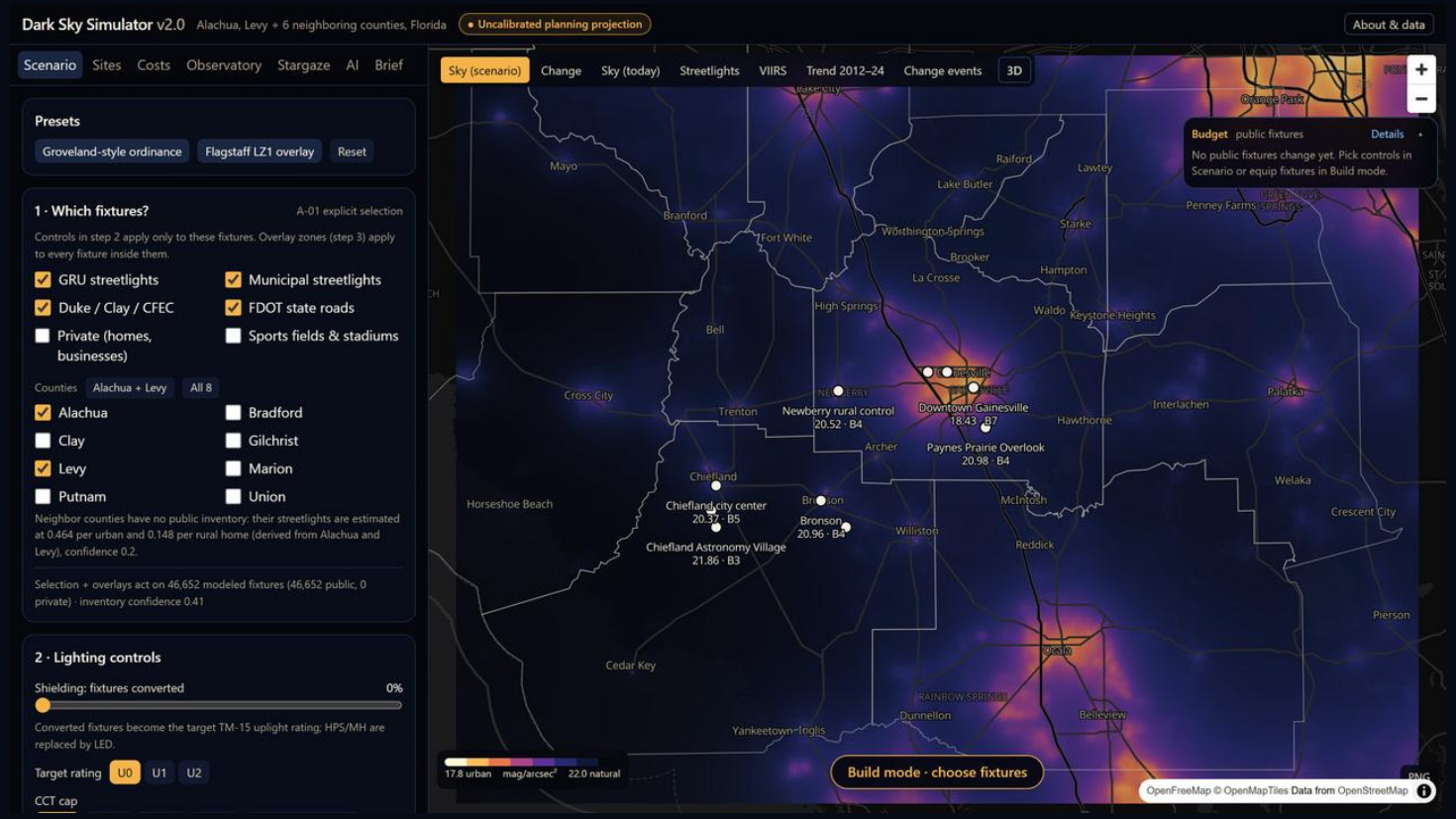
v2 · 2026

- Hand-tuned multipliers, built in a month
- Persuasive headline numbers (“$1.9M a year saved”) that nobody could audit

- Physics for the sky, public data for the lights, **AI only where data can teach**
- Every number labeled a planning projection; runs on a phone, no server

v2 at a glance

- **8 counties**: Alachua, Levy and six neighbors, ~0.4 km cells
- **~817,000** modeled light sources: streetlights, homes and businesses, sports fields
- Map: modeled zenith sky brightness (mag/arcsec^2); yellow = urban, black = near natural
- Seven tabs: Scenario · Sites · Costs · Observatory · Stargaze · **AI** · Brief



Scenario tab with today's modeled sky. Amber badge on every screen: “Uncalibrated planning projection”.

HOW IT WORKS

Heavy work offline, instant scenarios in the browser

Public data	Offline pipeline · Python	Static data package	Browser · React + MapLibre
NASA VIIRS night lights (Google Earth Engine) Census · OpenStreetMap streetlight inventories parcels · elevation Globe at Night meters	Atmospheric scattering kernels (Garstang) place every fixture AI: change detector + LightGBM forecast	16-bit grids + JSON one layer per fixture group × county (basis layers)	sky = weighted sum of basis layers ~55 ms per scenario map · 3D · charts · PDF

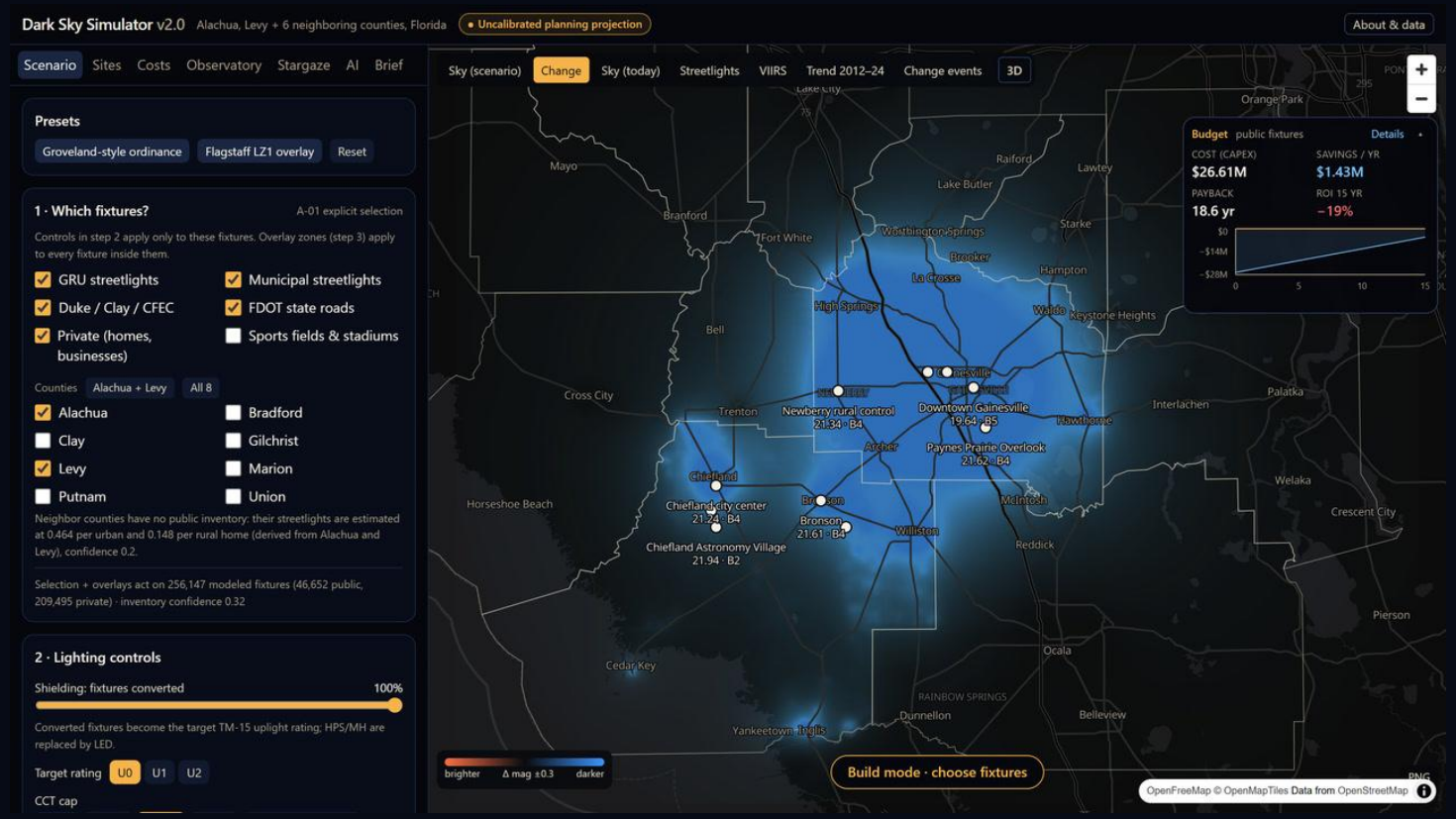
$$L^{\mathrm{scn}}(x) = \alpha \sum_{c,\,t} \Phi^{\mathrm{scn}}_{c,t}\, B_{c,t}(x), \qquad m = 12.6 - 2.5 \log_{10}\left(L_{\mathrm{art}} + L_{\mathrm{nat}}\right)$$

- Skyglow adds linearly, so any ordinance is a re-weighting of precomputed layers: a phone can explore it instantly
- The sky is a published single-scattering model (Garstang 1986), **not** a neural network

SCENARIO

One click: a Groveland-style ordinance

- Groveland: Florida's first dark-sky city
- Full shielding (U0), **3000 K** cap, dim to 50% from 22:00 to 06:30
- Paynes Prairie: **20.98 → 21.62** mag/arcsec^2 if every light complies; **21.30** with public lights alone
- Blue = darker sky, and it crosses county lines
- Public cost: **$26.6M**, $1.43M/yr saved, 18.6-yr payback



Change layer: where the sky gets darker (blue) under the ordinance; Budget card at top right.

SCENARIO · COSTS

What each government pays, and saves

Right-sized public streetlights: **fully shielded 3000 K LEDs at 30% lower output**, no smart controllers. Savings = energy + avoided maintenance, at each utility's rates.

Alachua County

all public streetlights (43,517)

$15.3M	$2.55M	6.0 yr	+$23.0M
one-time cost	saved / year	payback	after 15 years

University of Florida*

4,100 campus pole lights

$984K	$198K	5.0 yr	+$2.0M
one-time cost	saved / year	payback	after 15 years

City of Gainesville

GRU streetlights (30,000)

$7.20M	$1.45M	5.0 yr	+$14.6M
one-time cost	saved / year	payback	after 15 years

State roads (FDOT), Alachua

1,500 state-road lights

$900K	$113K	8.0 yr	+$795K
one-time cost	saved / year	payback	after 15 years

Levy County

all public streetlights (3,135)

$1.88M	$228K	8.3 yr	+$1.54M
one-time cost	saved / year	payback	after 15 years

City of Chiefland

285 city streetlights

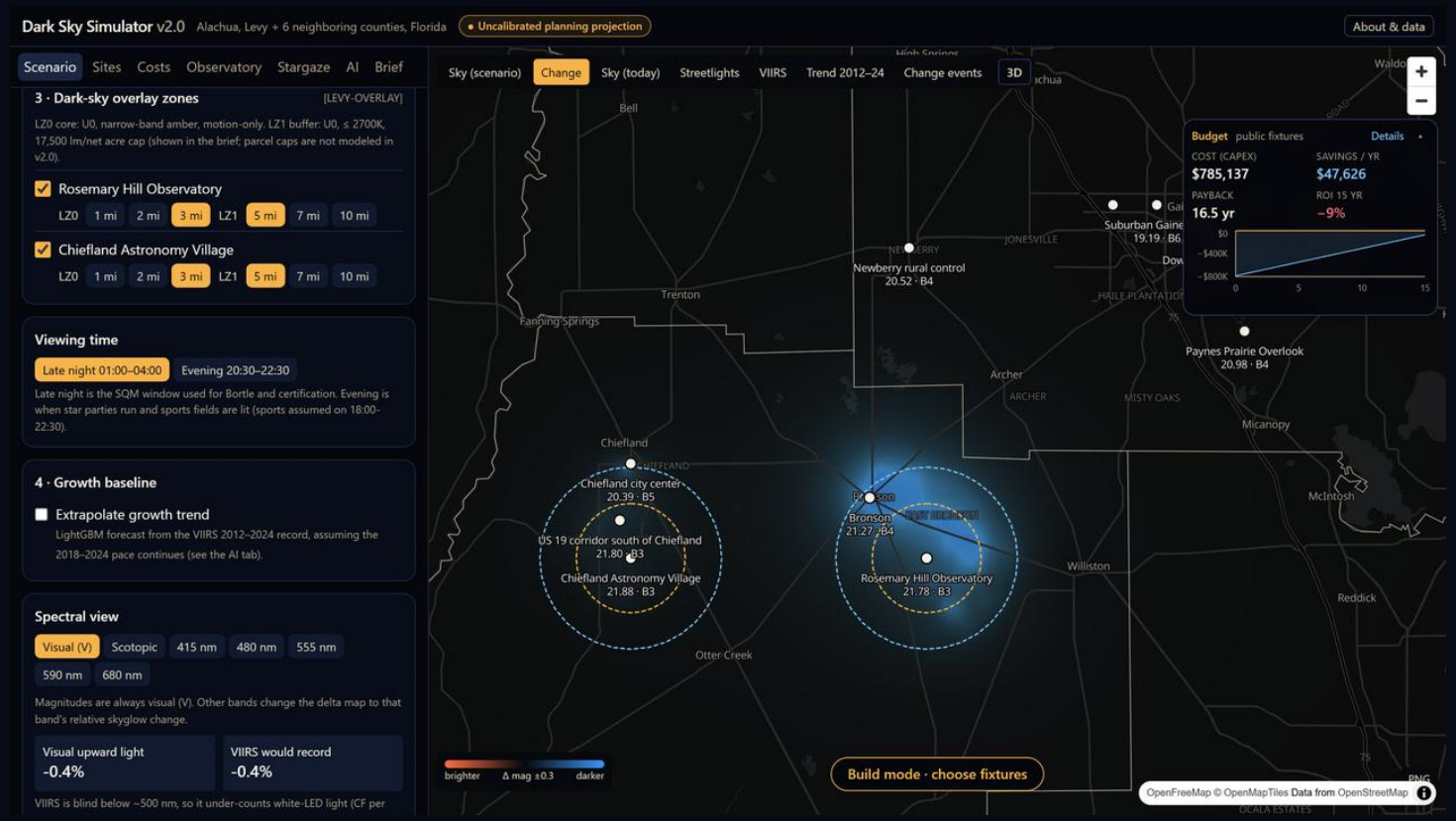
$171K	$21.5K	8.0 yr	+$152K
one-time cost	saved / year	payback	after 15 years

*UF: 4,100 campus pole lights, about 60% already LED (UF News, 2023), run through the simulator's cost engine at GRU rates. Gainesville and FDOT are within Alachua's total.

SCENARIO

Or protect just the observatories

- Overlay zones: stricter rules in rings around **Rosemary Hill** and **Chiefland Astronomy Village**
- LZ0 core: shielded, amber, motion-only
- LZ1 buffer: shielded, ≤ 2700 K
- Flagstaff-style preset: about **$0.79M** in public retrofits
- A trade-off a commission can actually vote on



Flagstaff LZ1 overlay preset (3-mile core, 5-mile buffer) on the Change layer.

SEE IT

Paynes Prairie, today and with the ordinance



Today



With a Groveland-style ordinance (every light, public and private, complies)

All-sky view from Paynes Prairie Overlook, looking southeast as the Milky Way rises: May 28, 2022, from 1:30 AM, no Moon. Illustrative, not a photograph.

- **20.98** mag/arcsec^2 · Bortle 4
- Faintest naked-eye stars ≈ magnitude 6.1

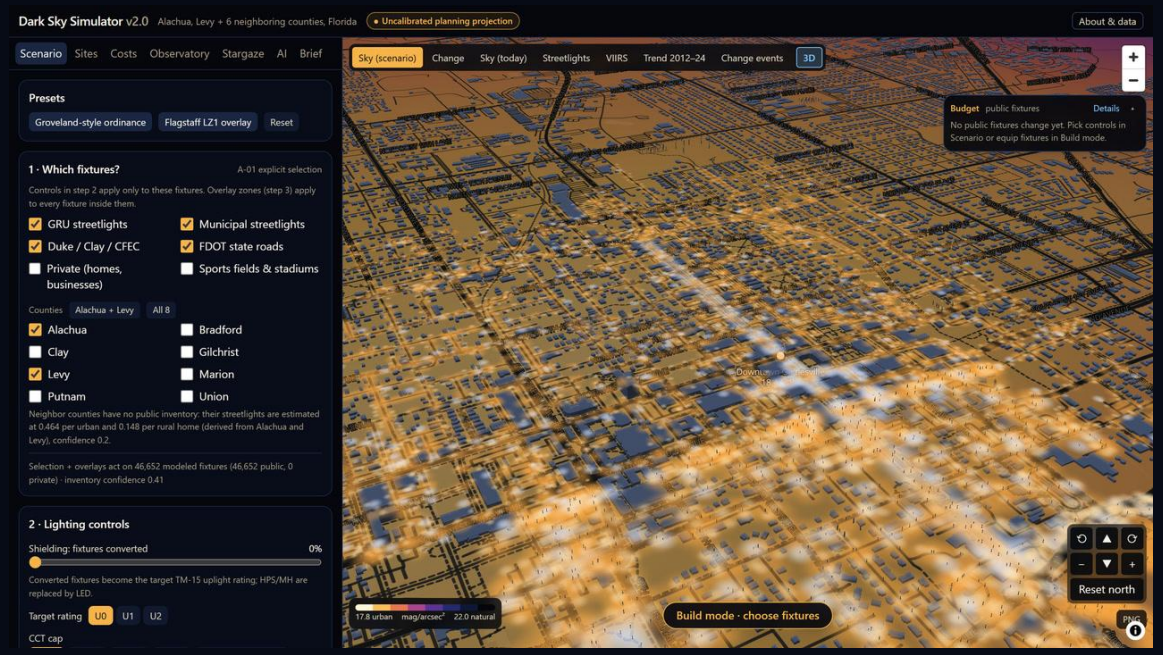
- **21.62** mag/arcsec^2 (+0.64) · still Bortle 4
- Faintest stars ≈ 6.4: a clearer Milky Way

SCENARIO

Build mode and 3D streets



Build mode



3D: downtown Gainesville

- Catalog of **17 DarkSky-compliant fixture types**: spectrum, uplight rating, watts, installed cost
- Equip one: sky, budget, brief and install markers update

- Terrain, buildings and every modeled light at street level
- Mapped streetlights where they stand; modeled ones along the roads of their cell

Where AI is used, and where it isn't

1 · Data
NASA Black Marble
13 years, 2012–2024

→

2 · Cleaning
remove the
solar-cycle airglow

→

3 · Statistics
step test on every
lit cell → change events

→

4 · Machine learning
LightGBM forecast
of where light grows

→ change-events map

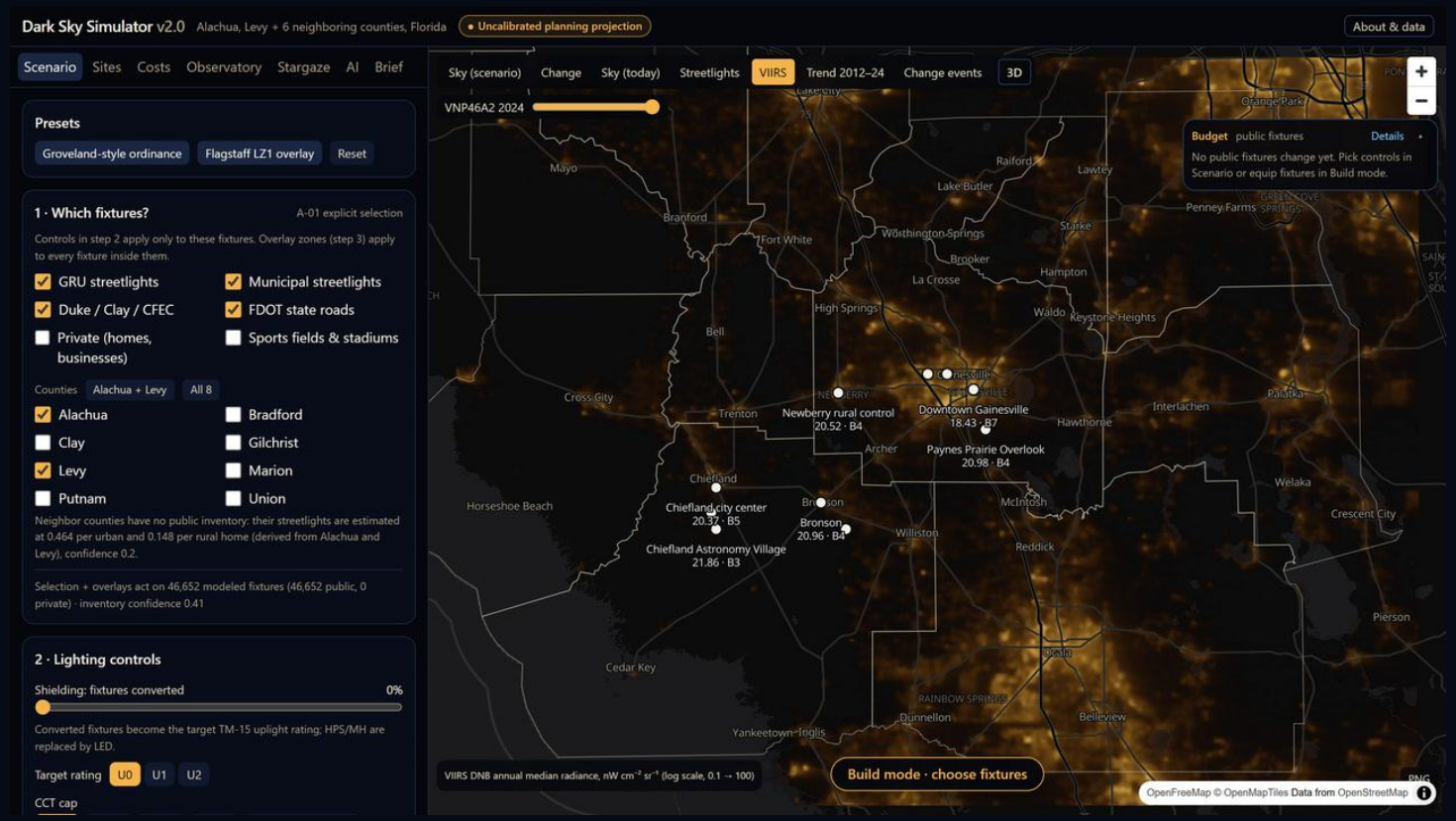
→ 2034 sky and observatory site ranking

NOT AI: physics and explicit rules you can inspect

- **Sky brightness**: atmospheric scattering physics
- **Scenario effects**: published fixture rules
- **Site ranking**: weighted sum, weights you set
- **Costs**: unit prices × fixture counts

The data: 13 years of NASA night lights

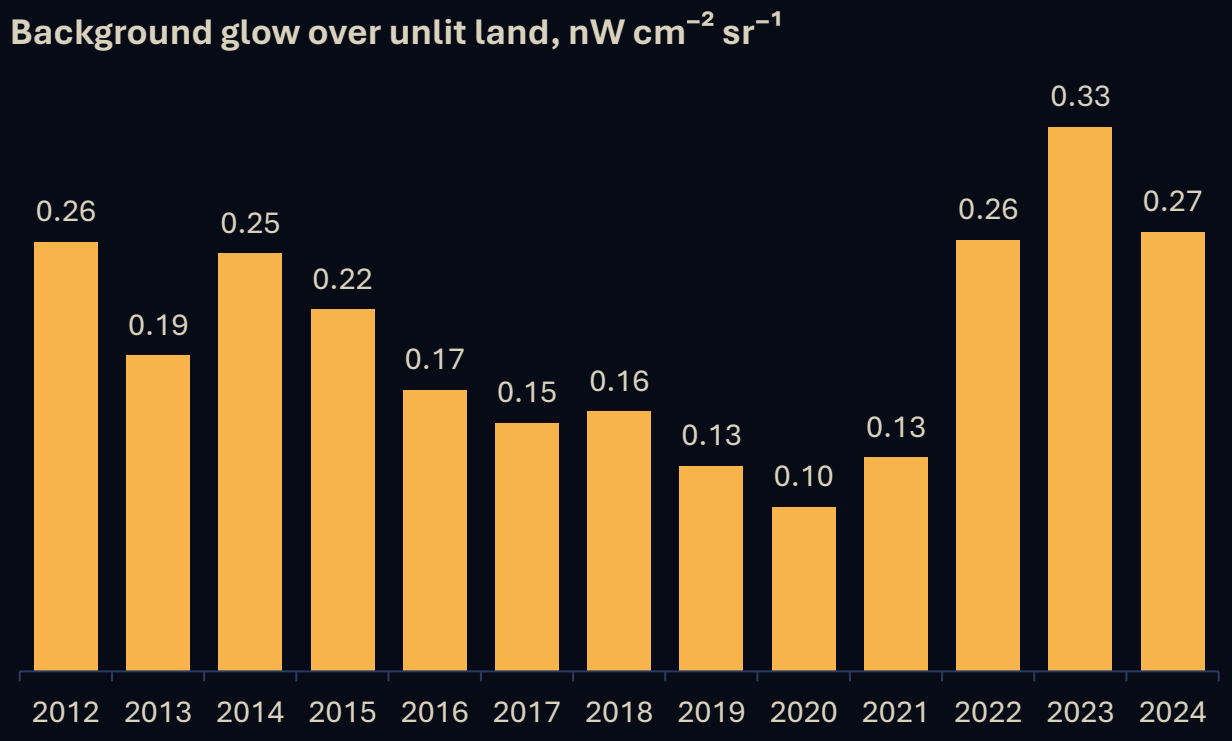
- NASA Black Marble (**VIIRS VNP46A2**): nightly, moonlight- and angle-corrected, ~500 m
- Keep cloud-free, high-quality nights; **one median per year** per cell, 2012–2024
- Pulled through **Google Earth Engine**'s free noncommercial tier in ~12 minutes
- The 2024 layer also tells the physics model where private lighting is



VIIRS layer: yearly median radiance with the airglow background removed; the slider steps through 2012–2024.

Lesson 1: check the data before the model

- First pull came back **10× too dark**: a scale factor applied twice
- Then forests and wetlands “brightened” **70–120%** from 2020 to 2023
- Cause: **airglow**, upper-atmosphere light that follows the 11-year solar cycle
- Fix: subtract each year's background, the median of the darkest fifth of the region, before any model sees the data



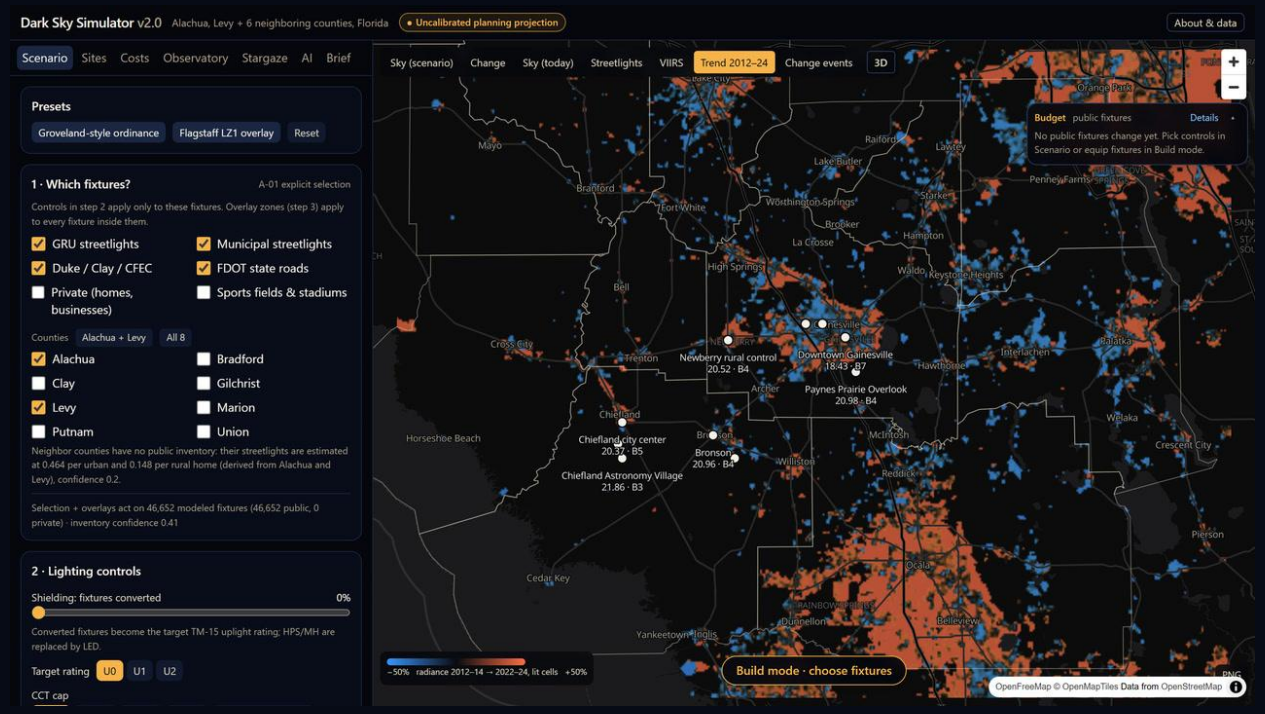
Solar minimum 2019–20: 0.10 · rising solar activity 2023: 0.33

$$\tilde{L}_t(x) = \max\left(L_t(x) - b_t,\ 0\right), \qquad b_t = \operatorname{median}_{x \in \mathcal{D}} L_t(x)$$

Revisiting the 19% estimate with calibrated satellite data

- AI analysis of NOAA night imagery, 2012 → 2024: **≈19% brighter** (v1)
- Re-measured in calibrated radiance, 2013–24: the **region around Alachua County rose ~20%** (NOAA) or **~13%** (NASA)
- Inside the county line: **3–6%**; around Gainesville ≈5%. Growth concentrates along I-75 and around Ocala

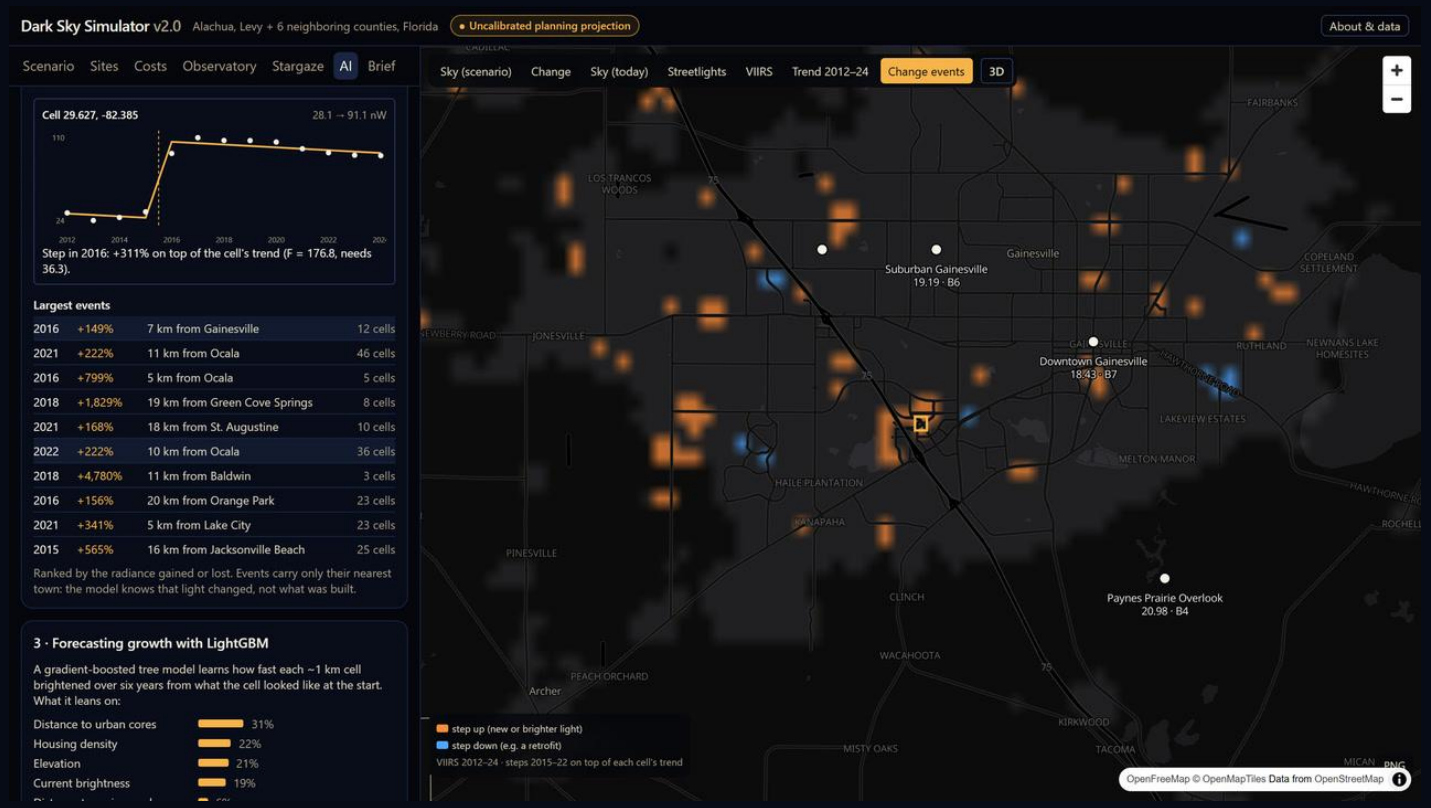
Calibrated radiance, 2013 → 2024	NOAA EOG	NASA BM
Region around the county	**+20.5%**	**+12.8%**
Alachua County	+6.1%	+2.9%
Gainesville	+5.0%	+4.8%



Trend 2012–24 layer: change from 2012–14 to 2022–24 for lit cells; growth along I-75 and around Ocala.

It found new development on its own

- **28,986** lit cells tested for a sudden step · **1,493** flagged as candidates · ≈ 29 expected by chance
- #1 event, near I-75 and Archer Road: **28 → 91 nW**, step in 2016 (F = 176.8)
- That's **Celebration Pointe**: Bass Pro Shops opened there in November 2016
- #2: dark land west of Ocala lit up in 2021 (0.5 → 53.5 nW)
- Nobody told the model where to look; it knows light changed, not what was built

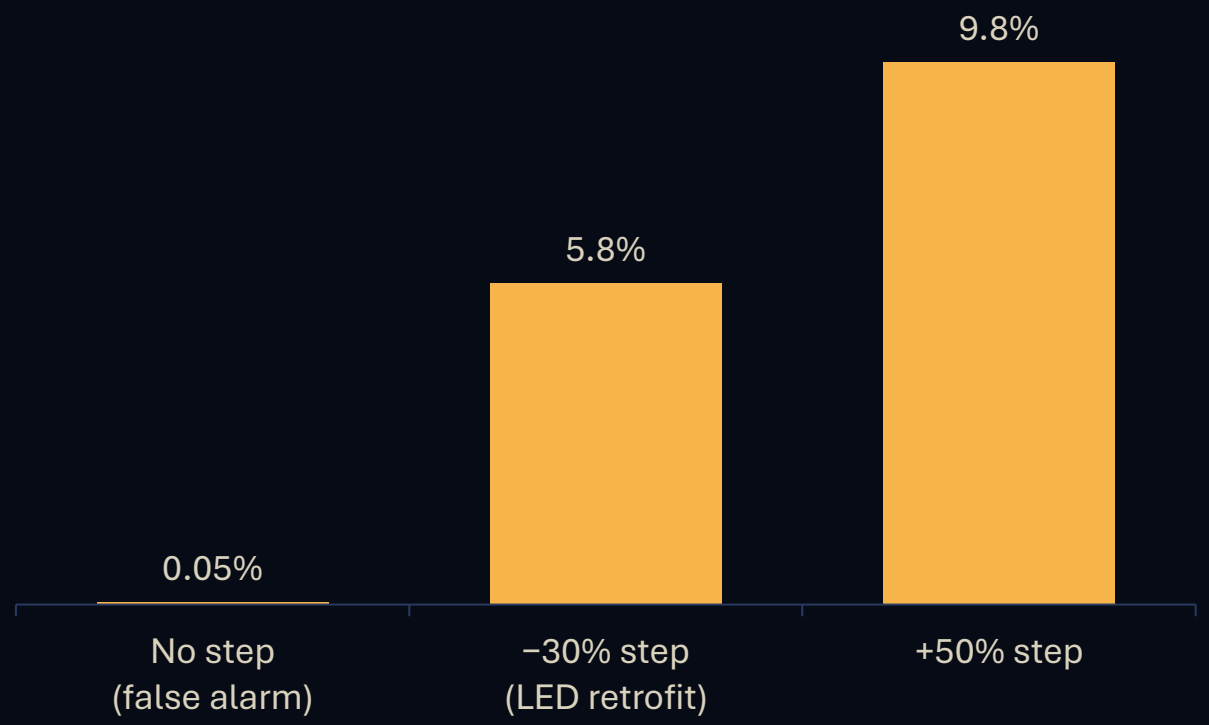


AI tab: clicking the top event flies the map to it and shows the cell's 13 years with the fitted step.

What the detector can't see

- A lit cell's yearly values scatter **≈ 10%** around its trend
- At that noise a +50% step is caught **10%** of the time, a −30% step **6%**
- So it finds large new lighting and **misses most LED retrofits** (20–40% drops)
- VIIRS is nearly blind below 500 nm: it misses most of a white LED's blue
- Next: monthly data and utility work orders

Chance a step is flagged, at the median noise (9.7%)



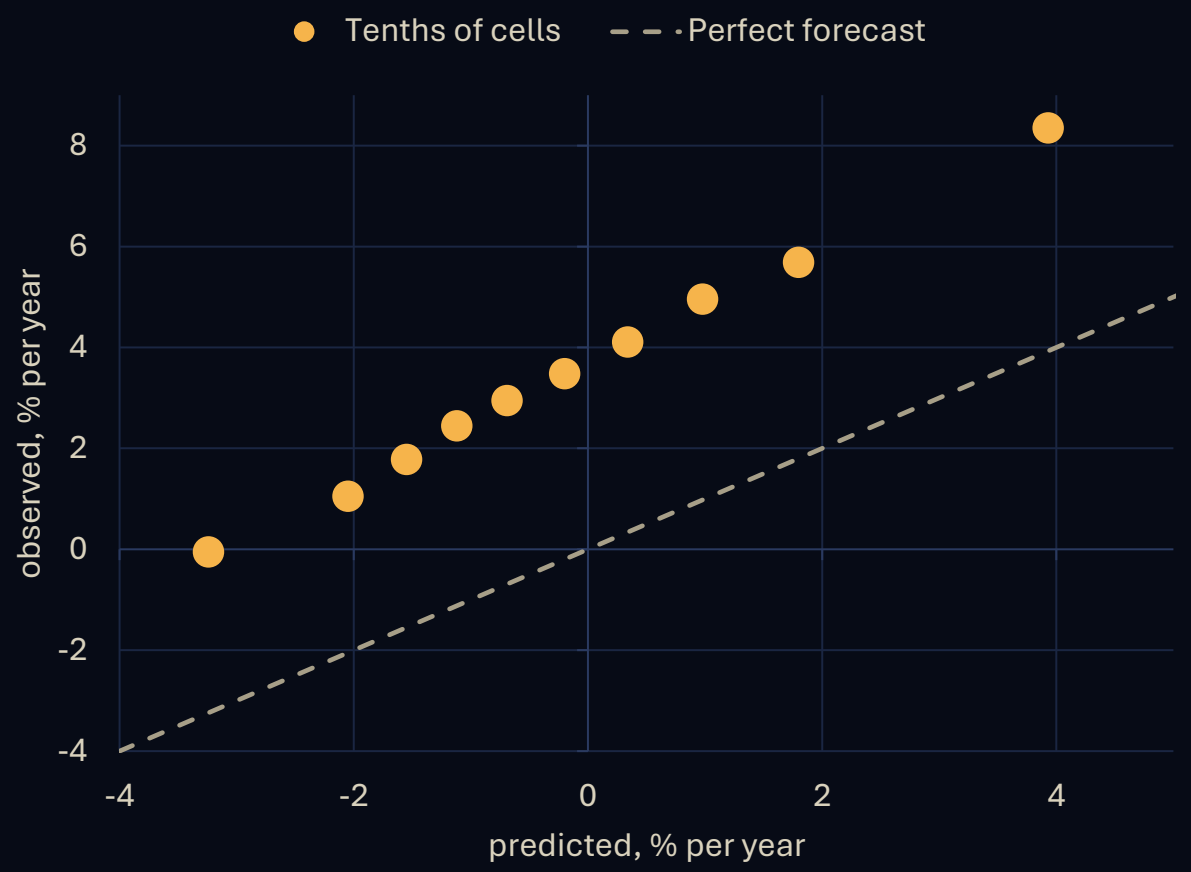
From 20,000 simulated series per case with a step in 2019. Large new development (+100% and more) is what it reliably finds.

The back-test: did it beat “no change”?

- Trained on 2012 → 2018, asked for **2024**: years it never saw
- Ties “no change” on brightness: lit areas fell **3.7%** (2012–18), then rose **20.6%** (2018–24)
- It ranks cells modestly well: correlation **0.35**; R^2 **0.18** on held-out areas

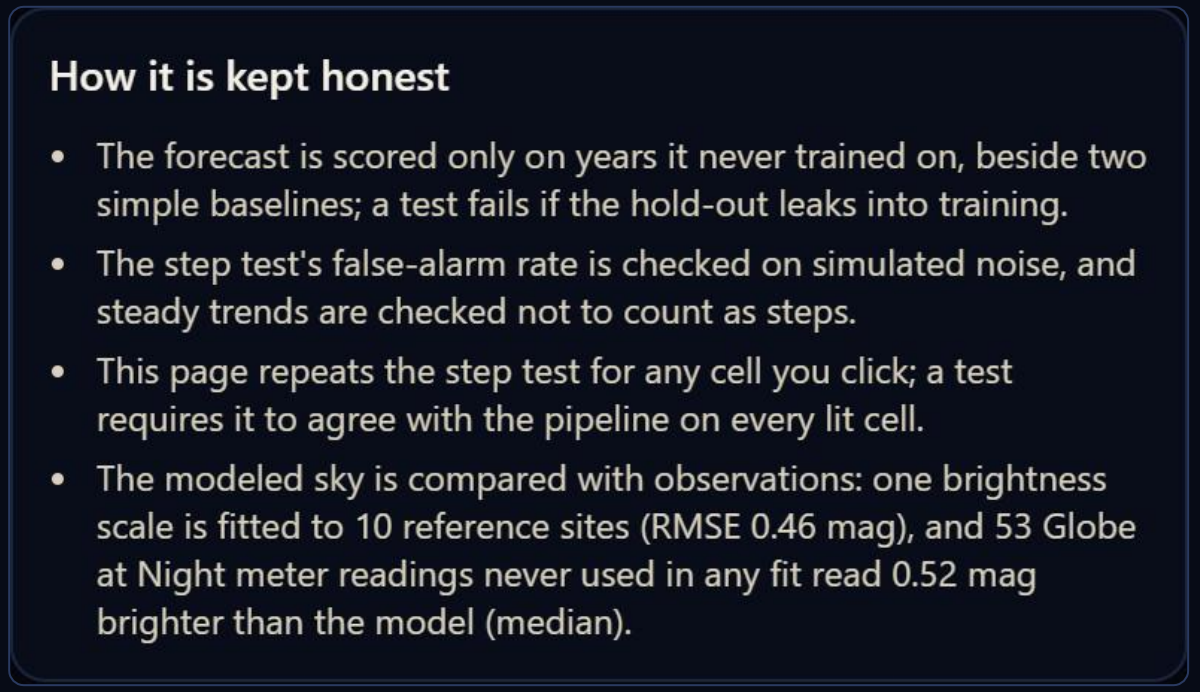
Prediction of 2024	Median error	Log error
LightGBM	**21.2%**	**0.449**
No change (2024 = 2018)	21.2%	0.467
Each cell's own past trend	25.4%	–

Predicted vs observed growth, 2018–24, by tenth of cells



How the AI is kept honest

- Forecast scored only on years it never trained on; a **unit test fails** if they leak into training
- Detector's false-alarm rate checked on simulated noise; steady trends must not count as steps
- The web page **re-runs the step test** for any cell you click; it must match the pipeline on every lit cell
- Sky model vs **53** Globe at Night meter readings never used in any fit: median 0.52 mag



How it is kept honest

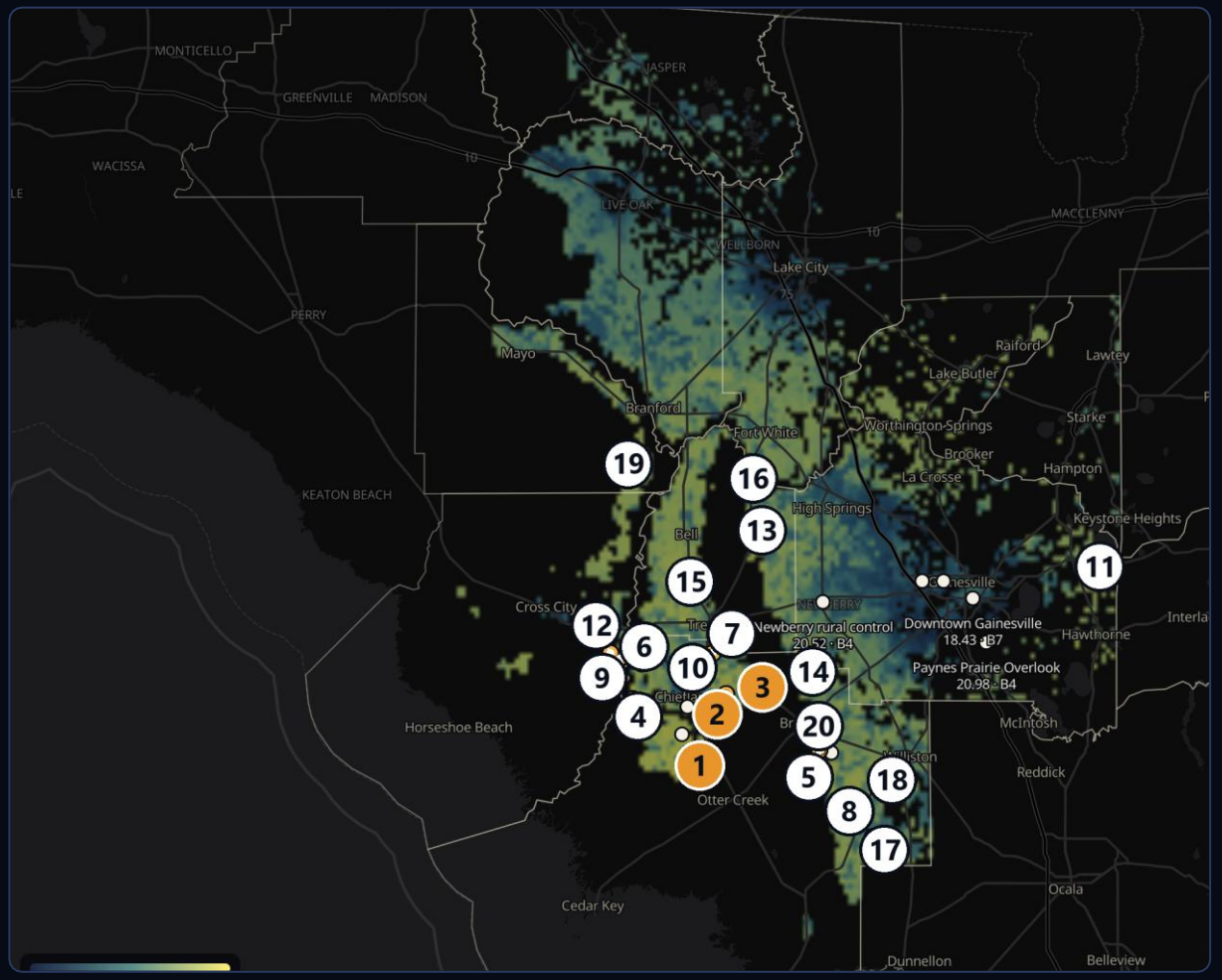
- The forecast is scored only on years it never trained on, beside two simple baselines; a test fails if the hold-out leaks into training.
- The step test's false-alarm rate is checked on simulated noise, and steady trends are checked not to count as steps.
- This page repeats the step test for any cell you click; a test requires it to agree with the pipeline on every lit cell.
- The modeled sky is compared with observations: one brightness scale is fitted to 10 reference sites (RMSE 0.46 mag), and 53 Globe at Night meter readings never used in any fit read 0.52 mag brighter than the model (median).

The same checks, stated in the app's AI tab.

IN USE

Where could UF build its next observatory?

- Requested by **Dr. Elizabeth Lada**, Chair of UF Astronomy
- Every ~1 km cell of 11 counties scored: dark sky today and in **2034** (LightGBM), clear air, land, access, **drive time from UF**
- Ruled out: **water and wetland**, the coast, ground below 5 m (storm surge), **2080 flood projections**, **FEMA's 100-year floodplain**, over ~90 min away
- **Shortlist: Levy #1–#3**, between Chiefland and Bronson: 44–58 min from campus, Bortle 3, land **$605K–806K** for 80 acres
- Telescope + building: $1–3M in published examples → **≈ $1.6–3.8M** total

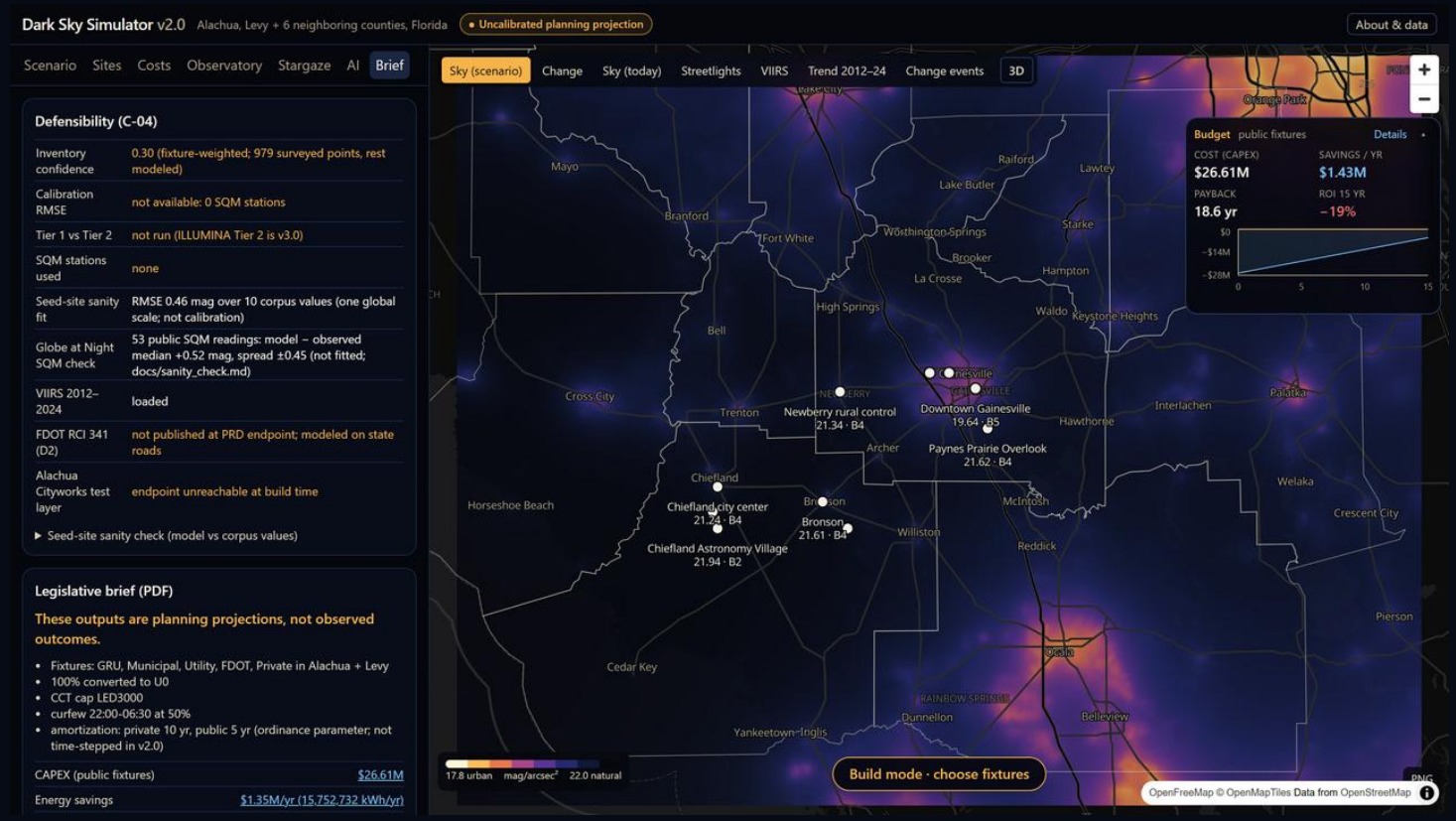


Observatory tab: the top 20 sites by rank (orange = Levy shortlist; markers enlarged). Blank areas are ruled out: water, wetland, flood-prone ground or over 90 minutes away.

IN USE

From model to policy brief

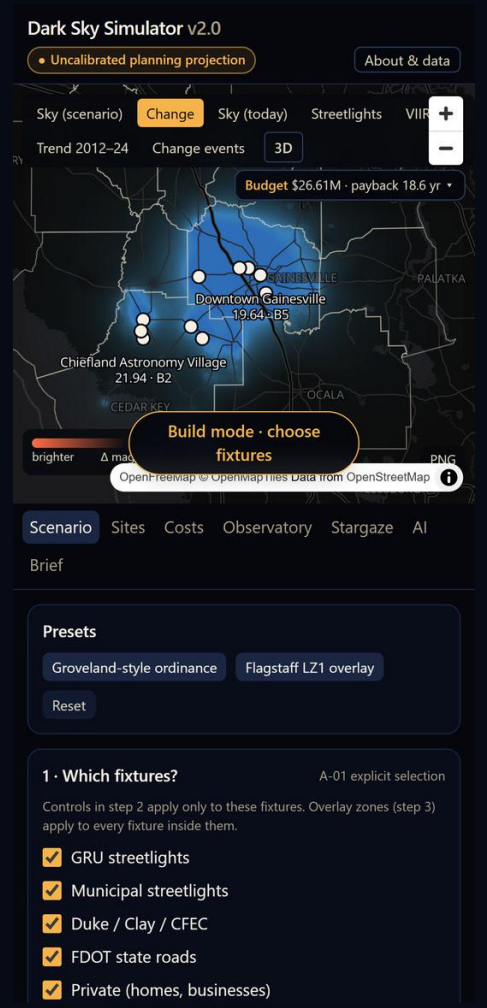
- One-click **PDF** for commissioners; every figure links back to its scenario
- First line: “These outputs are planning projections, not observed outcomes.”
- Defensibility panel lists what's missing: sky-meter stations, utility inventories, satellite calibration
- That list is the **v3 plan**



Brief tab: defensibility panel and the legislative brief.

Scaling up: from projection to evidence

- **Ground truth**: sky-quality meters in every county + a 50-node low-cost sensor pilot (~$5,900 hardware)
- **Real inventories**: utility data-sharing agreements (GRU, Duke, Clay, CFEC)
- **Computer vision**: a YOLO detector on street-level imagery to classify fixtures
- **Retrofit detection**: monthly satellite data; fit the LED correction on GRU's 3000 K conversions
- **Scaling**: a reusable county package (satellite data, lighting, development, policy), Levy County next
- **Siting**: parcel-level flood maps and site visits; blind test with UF Astronomy
- Looking for collaborators in sensors, computer vision, remote sensing and policy



The same app on a phone

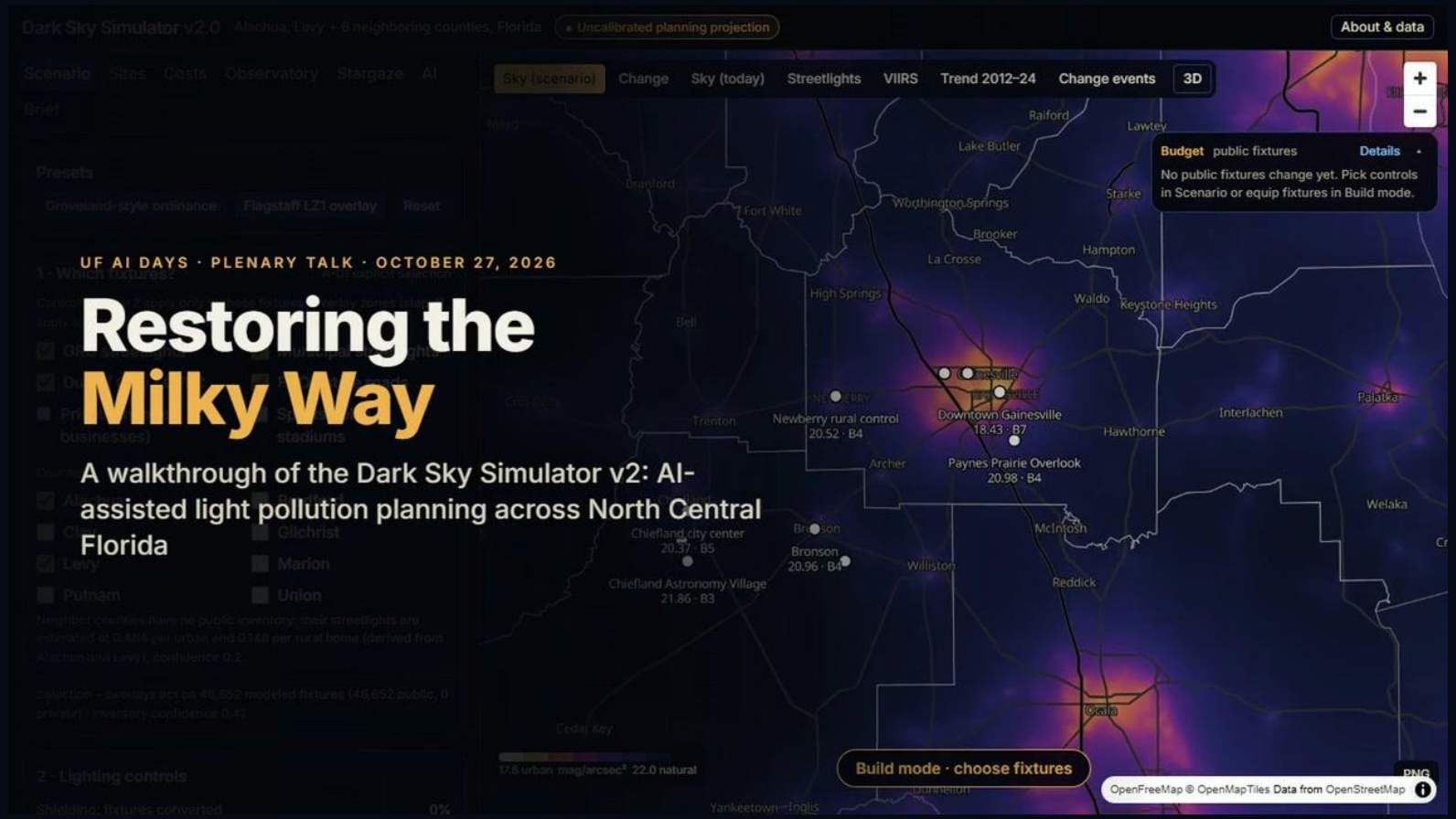
WHAT'S NEXT

What UF can do

- Alachua County has **adopted** dark-sky principles; **UF plans its own campus**, coordinated with the county, so it can lead
- **Lead on campus**: upgrading UF's 4,100 pole lights costs about **$984K** and saves about **$198K** a year
- **Protect UF's skies**: keep Rosemary Hill dark and choose a site for UF Astronomy's next observatory
- **Measure the real sky**: a 50-sensor network (about **$6,000** in hardware) to check the model against real readings
- Independent and unfunded so far: I'd like this to become a UF project



UF's Rosemary Hill Observatory. Photo: The Gainesville Sun, April 2012.



Thank you

Questions?

Try it

dark-sky-simulator-v2.vercel.app/dark-sky

Code

github.com/jayrosen-design/dark-sky-simulator-v2

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